Nimatullah

Computational Geophysics Researcher | PhD Scholar, IIT Bombay

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Profile

Computational geophysics researcher specializing in inverse problems, scientific machine learning, and high-performance computing for gravity, magnetic, and electromagnetic methods. My research integrates physics-guided deep learning, numerical optimization, and GPU-accelerated scientific computing to develop scalable inversion algorithms for large-scale subsurface imaging.

Education

Indian Institute of Technology Bombay *Jul 2024 – present*
PhD, Geophysics
CGPA: **9.26** / 10
Supervisors: Prof. Anand Singh and Dr. Pankaj K. Mishra.

Indian Institute of Technology Bombay *Aug 2022 – Aug 2024*
MSc, Applied Geophysics
CGPA: 7.76 / 10
Coursework included Electromagnetic Methods, Numerical Modelling, Signal Processing, Applied Geophysics, MATLAB. MSc dissertation: Modelling and Inversion of Cross-Well Electrical Resistivity Tomography (ERT).

Jamia Millia Islamia, New Delhi *2019 – 2022*
BSc, Physics
CGPA: 8.45 / 10
Coursework included Mathematical Physics, Electromagnetism, Classical Mechanics, and Thermodynamics.

Research & Professional Experience

Visiting Researcher | Geological Survey of Finland (GTK), Espoo, Finland *Dec 2025 – Jan 2026*

- Developed a high-performance **3-D gravity inversion framework in Julia** with data-space inversion strategies and GPU acceleration under the MULTIVERSE project.
- Implemented backend-agnostic computational kernels for seamless execution on CPU and GPU architectures; demonstrated order-of-magnitude speedups on million-cell models.
- Collaborated with researchers at the Geological Survey of Finland (GTK) on the EU MULTIVERSE project to develop scalable GPU-accelerated gravity inversion algorithms.

PhD Researcher | Dept. of Earth Sciences, IIT Bombay *Jul 2024 – present*

- Built a modular **3-D gravity inversion framework in Julia** with conjugate-gradient optimisation, Tikhonov regularisation, and depth weighting; validated on field datasets (manuscript under review).
- Developed a **magnetic inversion U-Net** to recover 2-D susceptibility from total-field anomaly grids.
- Developed Physics-Guided Neural Networks for basement depth estimation from gravity data, integrating physical constraints with deep learning.
- Developed Bayesian inversion workflows for transient electromagnetic data using conditional normalizing flows and neural posterior estimation.
- TA for GS 543 (Computer Programming for Geosciences), GP 414 (Electrical Methods), and GP 505 (Electromagnetic Methods); supported labs, graded assignments, and mentored undergraduates.

Summer Intern | Dept. of Earth Sciences, IIT Bombay *May – Jun 2023*

- Executed magnetic and VLF-EM surveys for groundwater exploration, including acquisition, processing, inversion modelling, and geological interpretation.

Publications

[1] **Basement depth estimation from gravity data using Physics-Guided Neural Networks and its comparison with data-driven deep learning** — Published

Kuldeep Sarkar, Nimatullah, Anand Singh, and Upendra K. Singh | Computers & Geosciences, 214, 106200 | 2026

A Physics-Guided Neural Network using a modified Xception encoder–decoder and a physics-constrained loss for basement-depth estimation from gravity anomalies.

DOI: 10.1016/j.cageo.2026.106200

[2] **Backend-agnostic Julia framework for 3D modeling and inversion of gravity data** — Under Review (Preprint on arXiv)

Nimatullah, Pankaj K. Mishra, Jochen Kamm, Anand Singh | arXiv:2602.03857 | 2026

Backend-agnostic Julia framework for GPU-accelerated 3-D gravity forward modelling and data-space inversion using a unified CPU/GPU implementation.

DOI: 10.48550/arXiv.2602.03857

[3] **Physics-Guided Deep Learning for Enhanced 2-D Magnetic Inversion of Multi-Source Susceptibility Structures** — Under review

Nimatullah | Under review | 2026–present

A physics-guided deep-learning approach to 2-D magnetic inversion for multi-source susceptibility structures.

Field Experience

TEM	Loop-transmitter and receiver-coil deployment; multi-channel transient decay acquisition and conductivity-depth inversion.	*IIT Bombay*
DC Resistivity	Wenner and Schlumberger arrays, VES profiling, cross-well ERT, tomographic imaging, aquifer delineation.	*IIT Bombay*
VLF-EM	Tilt-angle and ellipticity profiling for shallow fault and fracture detection.	*IIT Bombay*
Ground Magnetics	TMI acquisition with proton-precession, diurnal correction, levelling, reduction to pole, and susceptibility inversion.	*IIT Bombay*

Technical Skills

- **Programming:** Python, Julia, MATLAB
- **Geophysics:** TEM forward modelling, 3-D gravity inversion, magnetic inversion, DC resistivity, VLF
- **Scientific Computing:** Sparse linear algebra, PCG/LSQR, Tikhonov regularisation, GPU acceleration, parallel computing
- **Machine Learning:** PyTorch, TensorFlow/Keras, U-Net, Physics-Guided Neural Networks
- **Software:** Git, Linux, HPC clusters, Jupyter Notebook, LaTeX

Certifications

Advanced Learning Algorithms — Coursera (DeepLearning.AI) — *Jul 2023*

Supervised Machine Learning: Regression & Classification — Coursera — *Jun 2023*

Programming for Everybody (Python) — Coursera — *Sep 2022*